

SOME SEQUENTIAL EQUIVALENCE TESTS THAT CONTROL BOTH THE SIZE AND POWER

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Summary: Suppose observations are taken from a normally distributed population with mean μ and variance σ^2 both unknown. The problem is to test $H_0 : |\mu| \geq \Delta$ against $H_a : |\mu| < \Delta$ for establishing equivalence, where the constant Δ is the limit of equivalence. Several new sequential procedures are proposed which control both the size and the power at $|\mu| = \gamma\Delta$ for a given $0 \leq \gamma < 1$. Their large sample properties on size and power are given. Monte Carlo simulations are used to assess and compare the small and moderate sample size performances of these new procedures and a previously available procedure. Some related equivalence problems are discussed briefly.

1. Testing for equivalence

Suppose observations $X_1, X_2, \dots$ are taken from a normally distributed population $N(\mu, \sigma^2)$ with mean μ and variance σ^2 both unknown. The basic problem is to design a test of

$$H_0 : |\mu| \geq \Delta \quad \text{against} \quad H_a : |\mu| < \Delta \quad (1.1)$$

which has, at least approximately, size α and power no less than β at $|\mu| \leq \gamma\Delta$, where $\alpha, \beta \in (0, 1)$, $\gamma \in [0, 1)$ and $\Delta > 0$ are four prescribed constants. The limits of equivalence are set at $|\mu| < \Delta$.

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Such a problem can arise in a bioequivalence study, in which it is required to demonstrate that μ , the difference on a pharmacokinetics measure between two different drugs (or two different formulations of the same drug), is sufficiently small, $|\mu| < \Delta$, so that the two drugs can be regarded as therapeutically equivalent. Chow and Liu (1992) provide a general introduction on this. Such a problem can also arise in more general equivalence studies where the purpose of a study is to demonstrate that the 'products' produced by two different 'production methods' are 'practically' the same. For example, in a child growth study, one question of interest is whether two different methods of measuring a child's height are comparable. So several children are measured by both the methods and the difference between the first and second methods on each child, X , can be assumed to have a normal distribution $N(\mu, \sigma^2)$. To establish the comparability of the two methods the hypotheses can be set up as in (1.1). The size α requirement ensures that the probability of falsely concluding equivalence is only a small number α , and therefore protects the public interest in the situation of a bioequivalence study. On the other hand, the power β requirement guarantees that with a probability of at least β the correct conclusion of equivalence will be reached when $|\mu| \leq \gamma\Delta$, and therefore the expensive drug can be replaced by the cheaper but therapeutically equivalent one to reduce the cost of the drug manufacturer.

There are two major statistical approaches to an equivalence study. The first one is the hypothesis testing approach, which tests the hypothesis in (1.1) at size α . The second one is the confidence interval approach, which is based on a $1 - \alpha$ level confidence interval for μ . If this confidence interval is within the limits of equivalence, i.e. contained in the interval $(-\Delta, \Delta)$, then equivalence is accepted; otherwise rejected. For the purpose of decision

making, either acceptance or rejection of equivalence, which is often the case in a bioequivalence study as argued fiercely by Westlake (1981), the two approaches are equivalent. See Hsu, Hwang, Liu and Ruberg (1994), Berger and Hsu (1996), and Bauer and Kieser (1996).

Let σ^2 be unknown and suppose a sample of n observations $X_1, \dots, X_n$ is taken from the $N(\mu, \sigma^2)$ population. Let $\bar{X}_n$ and σ_n^2 be the sample mean and sample variance respectively. Then Schuirmann's (1981, 1987) two one-sided tests procedure is to

$$\text{reject } H_0 \iff \sqrt{n}(\bar{X}_n - \Delta)/\sigma_n < -t_{n-1}^\alpha \text{ and } \sqrt{n}(\bar{X}_n + \Delta)/\sigma_n > t_{n-1}^\alpha$$

where t_{n-1}^α denotes the upper α quantile of a t -distribution with $n - 1$ degrees of freedom. This test is equivalent in terms of decision making to the symmetric $1 - \alpha$ level confidence interval I_S :

$$P\{|\mu| \in I_S \equiv [0, |\bar{X}_n| + t_{n-1}^\alpha \sigma_n / \sqrt{n}]\} \geq 1 - \alpha,$$

which was proposed by Liu (1990). Westlake's (1976) symmetric $1 - \alpha$ level confidence interval I_W always contains I_S , and so I_S is sharper than I_W (see Hsu et al., 1994). Anderson and Hauck (1983) proposed another test which is to

$$\text{reject } H_0 \iff \sqrt{n}|\bar{X}_n|/\sigma_n < c, \quad (1.2)$$

where c is the critical value. When the variance σ^2 is a known constant, Anderson and Hauck's (1983) test becomes

$$\text{rejecting } H_0 \iff \sqrt{n}|\bar{X}_n|/\sigma < R(\sqrt{n}\Delta/\sigma), \quad (1.3)$$

where the function $R(\cdot) = R_\alpha(\cdot)$ is determined uniquely by the equation

$$P\{|N(0, 1) + x| < R(x)\} = \alpha \quad (1.4)$$

where $N(0, 1)$ denotes a standard normal random variable. Test (1.3) is equivalent in terms of decision making to the symmetric $1 - \alpha$ level confidence

interval I_N :

$$P\{|\mu| \in I_N \equiv [0, \sigma R^{-1}(\sqrt{n}\bar{X}_n/\sigma)/\sqrt{n}]\} = 1 - \alpha$$

where $R^{-1}(\cdot)$ denotes the inverse function of $R(\cdot)$. Confidence interval I_N can in fact be deduced from a more general result of Bofinger, Hayter and Liu (1993). It is not difficult to show that I_N is always contained in $[0, |\bar{X}_n| + z^\alpha \sigma/\sqrt{n}]$, the interval I_S corresponding to the known σ^2 case, where z^α is the upper α point of the $N(0, 1)$ distribution. So, when σ^2 is a known constant, Anderson and Hauck's (1983) test is uniformly more powerful than Schuirmann's (1981, 1987) test. Recently Tseng (2002) shows that confidence interval I_N for the known σ^2 case has certain optimality properties. For the unknown σ^2 case, more details on the construction of uniformly more powerful tests than two one-sided tests procedure of Schuirmann can be found in Brown, Hwang and Munk (1997). The relationship between the confidence interval inverted from the family of two one-sided tests (for different Δ) and the limaçon (an interval having certain optimal volume property) is discussed in Brown, Casella and Hwang (1995).

The problem with the procedures mentioned above is that, when σ^2 is unknown, their sizes depend on the unknown variance σ^2 and so cannot be controlled at α . Furthermore, none of the aforementioned procedures considers the control of power. Indeed, no fixed sample size procedure will be able to achieve the control of size and power when the value of σ^2 is unknown.

In this paper sequential sampling is used to adapt Anderson and Hauck's test (1.3) for the known σ^2 case (which has been shown by Tseng (2002) to have certain optimality properties) to deal with the unknown σ^2 situation so that both the size and power are controlled closely as specified by (1.1). There are several sequential sampling schemes available in the literature. Ghosh, Mukhopadhyay and Sen (1997) give an excellent general review on sequential

sampling. We apply several of these sampling schemes to our problem to develop some new sequential equivalence tests. Previously, there is only one two-sample sequential procedure of Srinivasan and Langenberg (1986) that controls both the size and power. The large sample properties of the new procedures on size and power are provided in Section 2. Their performances for small and moderate sample sizes are assessed and compared by simulations in Section 3.

In many applications one-sided equivalence is of interest. The aim in this case is to show that one treatment is practically not inferior to another treatment. Another scenario of interest is the equivalence of two treatments in a parallel design. The formulations of hypothesis testing and some sequential procedures for these two equivalence problems are discussed briefly in Section 4.

2. Sequential procedures

Throughout this paper, the following notation is used: $\phi(\cdot)$ denotes the pdf of the $N(0, 1)$ distribution; $\langle x \rangle$ denotes the integer part of x ; m , a pre-specified integer larger than one, is the size of the pilot sample; $o(1/n_0)$ is a sequence of numbers satisfying $\lim_{n_0 \rightarrow \infty} n_0 o(1/n_0) = 0$.

Before stating the sequential procedures, let us look firstly at how to determine the sample size in the known σ^2 situation. If σ^2 had been known, we would take a sample of size

$$n_0 = n_0(\sigma) = (\delta\sigma/\Delta)^2 \quad (2.1)$$

and test the hypothesis by

$$\text{rejecting } H_0 \iff |\bar{X}_{n_0}| < \sigma R(\sqrt{n_0}\Delta/\sigma)/\sqrt{n_0} = \Delta R(\delta)/\delta, \quad (2.2)$$

where $\bar{X}_{n_0}$ is the sample mean, $R(\cdot)$ is the function defined by (1.4), and $\delta = \delta(\alpha, \beta, \gamma)$ is the unique solution of the equation

$$P\{ |N(0, 1) + \gamma\delta| < R(\delta) \} = \beta.$$

The values of δ and $R(\delta)$ are given in Table 1 for selected values of α, β and γ .

Since σ^2 is unknown, the sample size formula (2.1) cannot be used directly and it is necessary to employ a sequential sampling procedure to estimate the required sample size by using (2.1). Depending on how frequently σ and hence the sample size (2.1) is re-estimated while observations accrue, several sequential procedures can be defined.

The fully sequential procedure P_1 uses the fully sequential sampling scheme of Anscombe (1953) and Chow and Robbins (1965). It re-estimates σ and hence the sample size (2.1) after each new observation after a pilot sample, and works in the following way. Take a pilot sample of size m . Then take observations one by one until

$$T_1 = \inf\{ n \geq m : n > (1 + \xi_1/n)(\delta/\Delta)^2 \sigma_n^2 \}$$

where ξ_1 is a predetermined constant. On stopping sampling, the acceptance/rejection of H_0 is to mimic (2.2):

$$\text{reject } H_0 \iff |\bar{X}_{T_1}| < (1 + \eta_1/T_1)\Delta R(\delta)/\delta$$

where η_1 is a predetermined constant. The values of ξ_1 and η_1 are given below.

The mixed procedure $P_2(k, \lambda_1, \dots, \lambda_{k-2})$ uses the sampling scheme of Liu (1997a). It is a mixture of the batch sampling (see below) and fully sequential sampling, and operates in the following way. Suppose the integer $k (\geq 3)$ and constants $0 < \lambda_1 < \dots < \lambda_{k-2} < 1$ are predetermined. Take a pilot sample of size m , and take the next $k - 2$ samples sequentially with the size of the i th ($i = 2, \dots, k - 1$) sample being $M_{i-1} - M_{i-2}$ where $M_0 = m$

and

$$M_j = \max\{\langle \lambda_j (\delta/\Delta)^2 \sigma_{M_{j-1}}^2 \rangle + 1, M_{j-1}\}, \quad 1 \leq j \leq k-2.$$

Then take one observation at a time until

$$T_2 = \inf\{n \geq M_{k-2} : n > (1 + \xi_2/n)(\delta/\Delta)^2 \sigma_n^2\},$$

where ξ_2 is a predetermined constant. On stopping sampling, the acceptance/rejection of H_0 is again to mimic (2.2):

$$\text{reject } H_0 \iff |\bar{X}_{T_2}| < (1 + \eta_2/T_2)\Delta R(\delta)/\delta$$

where η_2 is a predetermined constant. The values of ξ_2 and η_2 are given below. The value of k determines that $k-1$ batch samples are taken before the one-at-a-time fully sequential sampling starts. The value of λ_i determines that, on average, $100\lambda_i\%$ of the total observations are taken in the first $i+1$ batch samples.

The k -sample procedure $P_3(k, \lambda_1, \dots, \lambda_{k-2})$ uses the sampling scheme of Hall (1981) and Liu (1997b). It re-estimates σ and hence the sample size (2.1) after each batch of new observations, and works in the following way. Suppose the integer $k(\geq 3)$ and constants $0 < \lambda_1 < \dots < \lambda_{k-2} < 1$ are predetermined. Take a first sample of size m , and then take the next $k-2$ samples sequentially with the size of the i th ($i = 2, \dots, k-1$) sample being $M_{i-1} - M_{i-2}$ where $M_0 = m$ and

$$M_j = \max\{\langle \lambda_j (\delta/\Delta)^2 \sigma_{M_{j-1}}^2 \rangle + 1, M_{j-1}\}, \quad 1 \leq j \leq k-2.$$

Finally, take a sample of size $T_3 - M_{k-2}$ where

$$T_3 = \max\{\langle (\delta/\Delta)^2 \sigma_{M_{k-2}}^2 + \xi_3 \rangle + 1, M_{k-2}\}$$

where ξ_3 is a predetermined constant. On stopping sampling,

$$\text{reject } H_0 \iff |\bar{X}_{T_3}| < (1 + \eta_3/T_3)\Delta R(\delta)/\delta$$

where η_3 is a predetermined constant.

The values of k and the λ_i 's in P_3 play similar roles to those in P_2 . The only difference between P_2 and P_3 is that the one-at-a-time fully sequential sampling in P_2 is replaced by one final batch sample.

The constants ξ_i and η_i are used for fine tuning and are determined from large sample results so that P_i has a size $\alpha + o(1/n_0)$ and a minimum power $\beta + o(1/n_0)$ over $|\mu| \leq \gamma\Delta$, $i = 1, 2, 3$. They are given by

$$\begin{aligned}\xi_1 &= \xi_2 = 2x_0 + 1.683, \quad \eta_1 = \eta_2 = y_0/R, \\ \xi_3 &= (4x_0 + 5 - \lambda_{k-2}) / (2\lambda_{k-2}), \quad \eta_3 = y_0 / (\lambda_{k-2}R)\end{aligned}$$

where $x_0 = x_0(\alpha, \beta, \gamma)$ and $y_0 = y_0(\alpha, \beta, \gamma)$ solve simultaneously $F(x_0, y_0) = 0$ and $G(x_0, y_0) = 0$, with

$$\begin{aligned}F(x, y) &= \phi(R + \delta) \left\{ y + (R + \delta)x - (R + \delta)^3/4 \right\} \\ &\quad + \phi(R - \delta) \left\{ y + (R - \delta)x - (R - \delta)^3/4 \right\}, \\ G(x, y) &= \phi(R + \gamma\delta) \left\{ y + (R + \gamma\delta)x - (R + \gamma\delta)^3/4 \right\} \\ &\quad + \phi(R - \gamma\delta) \left\{ y + (R - \gamma\delta)x - (R - \gamma\delta)^3/4 \right\}.\end{aligned}$$

The derivation of these is similar to the derivation in Liu (1997c), to which the reader is referred. Selected values of x_0 and y_0 are given in Table 1. Without fine tuning, i.e. setting $\xi_i = \eta_i = 0$, P_i has size $\alpha + O(1/n_0)$ and minimum power $\beta + O(1/n_0)$ over $|\mu| \leq \gamma\Delta$, $i = 1, 2, 3$. So the procedures with fine tuning achieve higher order accuracy on the size and power requirements.

Finally, the two-sample procedure of Srinivasan and Langenberg (1986), P_{SL} , uses the two-sample method of Stein (1945) and operates in the following way. First, take a pilot sample of size m and calculate

$$T_{SL} = \max \left\{ m, \left\langle \left\{ \sigma_m(t_{m-1}^\alpha + t_{m-1}^{(1-\beta)/2}) / (\Delta(1-\gamma)) \right\}^2 + 1 \right\rangle \right\}.$$

Then take a second sample of size $T_{SL} - m$ and, after observing the data,

$$\text{reject } H_0 \iff |\bar{X}_{T_{SL}}| < \Delta(\gamma t_{m-1}^\alpha + t_{m-1}^{(1-\beta)/2}) / (t_{m-1}^\alpha + t_{m-1}^{(1-\beta)/2}).$$

This procedure guarantees that the size and power requirements are satisfied. But its sample size may be substantially larger than those required by the three new procedures. In fact, the difference between the average sample size of this procedure and n_0 approaches infinity as $n_0 \rightarrow \infty$, which means the amount of oversampling is not bounded. This is a characteristic difference between this two-sample procedure and the three new procedures.

3. Results of a simulation study

To assess and compare the performances of P_1 , P_2 , P_3 and P_{SL} for small and moderate sample sizes, a simulation study based on 10,000 replications for each procedure has been carried out for the combinations of $\alpha = .10, .05, .01$, $\beta = .6(.1).9$, $\gamma = .00, .25, .50, .75$, $m = 10, 15$ and various values of n_0 . The procedures are P_1 , $P_2^A = P_2(3, .5)$, $P_2^B = P_2(4, .5, .8)$, $P_2^C = P_2(5, .5, .8, .96)$, $P_3^A = P_3(3, .5)$, $P_3^B = P_3(4, .5, .8)$, $P_3^C = P_3(5, .5, .8, .96)$ and P_{SL} . The details for $\alpha = .05$, $\gamma = .5$ and $\beta = .6, .9$ are reported here, while the results for other parameter combinations are similar. The results on size and power are given in Table 2, while the results on the average total sample size and standard deviation are given in Table 3.

To use any of the the sequential sampling procedures, we need to choose a value for m , the size of the pilot sample. In order to reduce the risk of oversampling, m should not be too large, since $m > n_0$ will cause outright oversampling. On the other hand, most studies require a certain number of observations, 10 say, in order for a study to be credible and for obvious deviation from the normality assumption to be detected. It is clear from Table 2 that, when $m = 10$ or 15, the sizes of $P_i(1 \leq i \leq 3)$ are very closely

controlled at the target level $\alpha = .05$, and the minimum powers are very close to the target power. From Table 3 it can be seen that the new procedures $P_i (1 \leq i \leq 3)$ require at most a few observations more than n_0 , the truly required sample size had σ^2 been known. But the total sample size of the two-sample procedure can be considerably larger than n_0 , and has a much larger variation. For example, when $m = 10$, $\beta = 0.6$ and $n_0 = 200$, the average sample size of P_{SL} is 407.4, more than twice of $n_0 = 200$. This problem is only slightly alleviated when the initial sample size m is increased from 10 to 15. This highlights the snag of oversampling inherent in the two-sample procedure.

Some previous work on the two-sample procedure (in a different context) suggests that if m is about twenty, then the risk of serious oversampling can largely be removed. This is actually not true; if $m = 20$ and $n_0 = 400$ say, the oversampling is still about 300. The only way to avoid serious oversampling for a two-sample procedure is to specify an m that is close to (and smaller than) the value of n_0 , which is impossible since n_0 is unknown.

Note that sampling operations also incur cost in real problems, and so it is relevant in choosing a procedure. In terms of the number of sampling operations required, the descending order is: P_1, P_2, P_3, P_{SL} . Since all the procedures control the type I and II error rates satisfactorily, one only needs to focus on the average sample size and the number of sampling operations in deciding which of the procedures should be used for a given problem. The following overall conclusions may be drawn. The two-sample procedure requires only two sampling operations, but has a risk of serious oversampling. It should therefore be employed only if the cost of an observation is negligible in comparison with the cost of one sampling operation. Otherwise, a suitable $k(\geq 3)$ -sample procedure P_3 or a mixed procedure P_2 should be used. By

increasing just one sampling operation, a three-sample procedure eliminates the risk of oversampling inherent in the two-sample procedure. Between a $k(\geq 3)$ -sample procedure P_3 and a mixed procedure P_2 , a $k(\geq 3)$ -sample procedure is preferable if the cost of sampling operation is more dominant than the cost of the observation. Otherwise a mixed procedure P_2 should be used. A fully sequential procedure P_1 should probably never be used since it requires almost the same average sample size as but a lot more sampling operations than a suitable mixed procedure P_2 . In a real situation, the precise decision on which one of the four procedures is the best can be made only if the costs of one observation and one sampling operation can be quantified.

4. Some related problems

The setting considered above is for the two-sided equivalence of one treatment to a given standard or of two treatments in paired comparison. In this section, we outline the theory for two related problems. The first is for the two-sided equivalence of two treatments in a parallel design. The second is for the one-sided equivalence of one treatment to a given standard or of two treatments in a paired comparison. Mixed and k -sample procedures only are developed for these two problems, as they are of most interest from the discussions in Section 3. Note that the key in developing the sequential procedures are the corresponding fixed sample size procedures for the known variance situation, such as those in (2.1) and (2.2).

4.1 Parallel design

The equivalence of two populations $N(v, \sigma^2)$ and $N(w, \sigma^2)$ is to be assessed by hypothesis testing:

$$H_0 : |\mu| = |v - w| \geq \Delta \quad \text{against} \quad H_a : |\mu| < \Delta$$

where the common variance σ^2 is an unknown parameter. So the two populations are deemed to be equivalent if the mean difference between the two populations is smaller than Δ . It is required that a test is to have at least approximately size α and power no less than β at $|\mu| \leq \gamma\Delta$ as before. Independent observations are available from the two populations: $X_1, X_2, \dots \sim N(v, \sigma^2)$, $Y_1, Y_2, \dots \sim N(w, \sigma^2)$. Let $\bar{X}_n, \bar{Y}_n$ be the sample means respectively, and let $\sigma_n^2 = (\sum_1^n [(X_i - \bar{X}_n)^2 + (Y_i - \bar{Y}_n)^2] / (2(n-1)))$ be the pooled sample variance.

If σ^2 had been known, we would take a sample from each population of size

$$n_0 = n_0(\sigma) = 2(\delta\sigma/\Delta)^2 \quad (4.1)$$

and test the hypothesis by

$$\text{rejecting } H_0 \iff |\bar{X}_{n_0} - \bar{Y}_{n_0}| < \sigma R(\sqrt{n_0/2}\Delta/\sigma)\sqrt{2/n_0} = \Delta R(\delta)/\delta, \quad (4.2)$$

where $R(\cdot)$ and $\delta = \delta(\alpha, \beta, \gamma)$ are defined in (1.4) and below (2.2) respectively.

Since σ^2 is unknown, (4.1) and (4.2) are used to construct sequential procedures. The mixed procedure $P_2(k, \lambda_1, \dots, \lambda_{k-2})$ operates in the following way. Suppose the integer $k(\geq 3)$ and constants $0 < \lambda_1 < \dots < \lambda_{k-2} < 1$ are predetermined as before. Take a pilot sample of size m from each population, and take the next $k-2$ samples from each population sequentially with the size of the i th ($i = 2, \dots, k-1$) sample being $M_{i-1} - M_{i-2}$ where $M_0 = m$ and

$$M_j = \max\{\langle 2\lambda_j(\delta/\Delta)^2\sigma_{M_{j-1}}^2 \rangle + 1, M_{j-1}\}, \quad 1 \leq j \leq k-2.$$

Then take one pair of observations (X, Y) at a time until

$$T_2 = \inf\{n \geq M_{k-2} : n > 2(1 + \xi_4/n)(\delta/\Delta)^2\sigma_n^2\},$$

where ξ_4 is a predetermined constant. On stopping sampling,

$$\text{reject } H_0 \iff |\bar{X}_{T_2} - \bar{Y}_{T_2}| < (1 + \eta_4/T_2)\Delta R(\delta)/\delta$$

where η_4 is a predetermined constant.

The size and minimum power of this procedure are given respectively by $\alpha + H_4(\xi_4, \eta_4)/n_0 + o(1/n_0)$ and $\beta + K_4(\xi_4, \eta_4)/n_0 + o(1/n_0)$, where

$$\begin{aligned} H_4(\xi_4, \eta_4) &= \phi(R+\delta) \left\{ \eta_4 R + (R+\delta)(4\xi_4 - 2.02)/8 - (R+\delta)^3/8 \right\} \\ &\quad + \phi(R-\delta) \left\{ \eta_4 R + (R-\delta)(4\xi_4 - 2.02)/8 - (R-\delta)^3/8 \right\}, \\ K_4(\xi_4, \eta_4) &= \phi(R+\gamma\delta) \left\{ \eta_4 R + (R+\gamma\delta)(4\xi_4 - 2.02)/8 - (R+\gamma\delta)^3/8 \right\} \\ &\quad + \phi(R-\gamma\delta) \left\{ \eta_4 R + (R-\gamma\delta)(4\xi_4 - 2.02)/8 - (R-\gamma\delta)^3/8 \right\}. \end{aligned}$$

The values of ξ_4 and η_4 can be solved from $H_4(\xi_4, \eta_4) = 0$ and $K_4(\xi_4, \eta_4) = 0$ so that the size is $\alpha + o(1/n_0)$ and the minimum power is $\beta + o(1/n_0)$.

The k -sample procedure $P_3(k, \lambda_1, \dots, \lambda_{k-2})$ works in the following way. Suppose the integer $k(\geq 3)$ and constants $0 < \lambda_1 < \dots < \lambda_{k-2} < 1$ are predetermined. Take a first sample of size m from each population, and then take the next $k-2$ samples from each population sequentially with the size of the i th ($i = 2, \dots, k-1$) sample being $M_{i-1} - M_{i-2}$ where $M_0 = m$ and

$$M_j = \max\{(\lambda_j(\delta/\Delta)^2 \sigma_{M_{j-1}}^2 + 1, M_{j-1}\}, \quad 1 \leq j \leq k-2.$$

Finally, take a sample of size $T_3 - M_{k-2}$ from each population where

$$T_3 = \max\{((\delta/\Delta)^2 \sigma_{M_{k-2}}^2 + \xi_5) + 1, M_{k-2}\}$$

where ξ_5 is a predetermined constant. On stopping sampling,

$$\text{reject } H_0 \iff |\bar{X}_{T_3} - \bar{Y}_{T_3}| < (1 + \eta_5/T_3)\Delta R(\delta)/\delta$$

where η_5 is a predetermined constant.

The size and minimum power of this procedure are given respectively by $\alpha + H_5(\xi_5, \eta_5)/n_0 + o(1/n_0)$ and $\beta + K_5(\xi_5, \eta_5)/n_0 + o(1/n_0)$, where

$$H_5(\xi_5, \eta_5) = \phi(R+\delta) \left\{ \eta_5 R + (R+\delta) (4\xi_5 + 2.5/\lambda) / 8 - (R+\delta)^3 / (8\lambda) \right\} \\ + \phi(R-\delta) \left\{ \eta_5 R + (R-\delta) (4\xi_5 + 2.5/\lambda) / 8 - (R-\delta)^3 / (8\lambda) \right\},$$

$$K_5(\xi_5, \eta_5) = \phi(R+\gamma\delta) \left\{ \eta_5 R + (R+\gamma\delta) (4\xi_5 + 2.0-5/\lambda) / 8 - (R+\gamma\delta)^3 / (8\lambda) \right\} \\ + \phi(R-\gamma\delta) \left\{ \eta_5 R + (R-\gamma\delta) (4\xi_5 + 2.0-5/\lambda) / 8 - (R-\gamma\delta)^3 / (8\lambda) \right\}.$$

The constants ξ_5 and η_5 are determined from $H_5(\xi_5, \eta_5) = 0$ and $K_5(\xi_5, \eta_5) = 0$ so that the procedure has a size $\alpha + o(1/n_0)$ and a minimum power $\beta + o(1/n_0)$.

4.2 One-sided equivalence

There are situations in which one wants to demonstrate that a treatment is no worse than a standard or one treatment is no worse than another treatment in paired comparison by amount Δ . Then the hypotheses should be set up as

$$H_0 : \mu \leq -\Delta \quad \text{against} \quad H_a : \mu > -\Delta.$$

Independent observations are $X_1, X_2, \dots \sim N(\mu, \sigma^2)$, where the variance σ^2 is an unknown parameter. It is required that a test is to have at least approximately size α and power no less than β at $\mu \geq -\gamma\Delta$. Let $\bar{X}_n$ be the sample mean and $\sigma_n^2 = \sum_1^n (X_i - \bar{X}_n)^2 / (n-1)$ be the sample variance.

If σ^2 had been known, we would take a sample of size

$$n_0 = n_0(\sigma) = \left(\frac{z_\alpha - z_\beta}{(1-\gamma)\Delta} \right)^2 \sigma^2 \quad (4.3)$$

and test the hypothesis by

$$\text{rejecting } H_0 \iff \bar{X}_n > -\Delta(1 - (1-\gamma)z_\alpha / (z_\alpha - z_\beta)) \quad (4.4)$$

where z_α, z_β are the upper α, β point of a standard normal distribution respectively.

Since σ^2 is unknown, (4.3) and (4.4) are used to construct sequential procedures. The mixed procedure $P_2(k, \lambda_1, \dots, \lambda_{k-2})$ operates in the following way. Suppose the integer $k(\geq 3)$ and constants $0 < \lambda_1 < \dots < \lambda_{k-2} < 1$ are predetermined as before. Take a pilot sample of size m , and take the next $k-2$ samples sequentially with the size of the i th ($i = 2, \dots, k-1$) sample being $M_{i-1} - M_{i-2}$ where $M_0 = m$ and

$$M_j = \max\left\{\left\langle \lambda_j \left(\frac{z_\alpha - z_\beta}{(1-\gamma)\Delta} \right)^2 \sigma_{M_{j-1}}^2 \right\rangle + 1, M_{j-1}\right\}, \quad 1 \leq j \leq k-2.$$

Then take one observation at a time until

$$T_2 = \inf\{n \geq M_{k-2} : n > (1 + \xi_6/n) \left(\frac{z_\alpha - z_\beta}{(1-\gamma)\Delta} \right)^2 \sigma_n^2\},$$

where ξ_6 is a predetermined constant. On stopping sampling,

$$\text{reject } H_0 \iff \bar{X}_{T_2} > -(1 + \eta_6/T_2)\Delta(1 - (1-\gamma)z_\alpha/(z_\alpha - z_\beta))$$

where η_6 is a predetermined constant.

The size and minimum power of this procedure are given respectively by

$$\alpha + z_\alpha \phi(z_\alpha) H_6(\xi_6, \eta_6)/n_0 + o(1/n_0)$$

and

$$\beta + z_\beta \phi(z_\beta) K_6(\xi_6, \eta_6)/n_0 + o(1/n_0),$$

where

$$H_6(\xi_6, \eta_6) = \left(\frac{\gamma z_\alpha - z_\beta}{(1-\gamma) z_\alpha} \right) \eta_6 + \frac{z_\alpha^2 - 2\xi_6 + 3.366}{4},$$

$$K_6(\xi_6, \eta_6) = \left(\frac{\gamma z_\alpha - z_\beta}{(1-\gamma) z_\beta} \right) \eta_6 + \frac{z_\beta^2 - 2\xi_6 + 3.366}{4}.$$

The values of ξ_4 and η_4 can be solved from $H_6(\xi_6, \eta_6) = 0$ and $K_6(\xi_6, \eta_6) = 0$ so that the size is $\alpha + o(1/n_0)$ and the minimum power is $\beta + o(1/n_0)$.

The k -sample procedure $\mathbf{P}_3(k, \lambda_1, \dots, \lambda_{k-2})$ works in the following way. Suppose the integer $k(\geq 3)$ and constants $0 < \lambda_1 < \dots < \lambda_{k-2} < 1$ are predetermined. Take a first sample of size m , and then take the next $k-2$ samples sequentially with the size of the i th ($i = 2, \dots, k-1$) sample being $M_{i-1} - M_{i-2}$ where $M_0 = m$ and

$$M_j = \max\left\{\left\langle \lambda_j \left(\frac{z_\alpha - z_\beta}{(1-\gamma)\Delta} \right)^2 \sigma_{M_{j-1}}^2 \right\rangle + 1, M_{j-1}\right\}, \quad 1 \leq j \leq k-2.$$

Finally, take a sample of size $T_3 - M_{k-2}$ where

$$T_3 = \max\left\{\left\langle \left(\frac{z_\alpha - z_\beta}{(1-\gamma)\Delta} \right)^2 \sigma_{M_{k-2}}^2 + \xi_7 \right\rangle + 1, M_{k-2}\right\}$$

where ξ_7 is a predetermined constant. On stopping sampling,

$$\text{reject } H_0 \iff \bar{X}_{T_3} > -(1 + \eta_7/T_3)\Delta(1 - (1 - \gamma)z_\alpha/(z_\alpha - z_\beta))$$

where η_7 is a predetermined constant.

The size and minimum power of this procedure are given respectively by

$$\alpha + z_\alpha \phi(z_\alpha) H_7(\xi_7, \eta_7)/n_0 + o(1/n_0)$$

and

$$\beta + z_\beta \phi(z_\beta) K_7(\xi_7, \eta_7)/n_0 + o(1/n_0),$$

where

$$\begin{aligned} H_7(\xi_7, \eta_7) &= \left(\frac{\gamma z_\alpha - z_\beta}{(1-\gamma) z_\alpha} \right) \eta_7 + \frac{z_\alpha^2 - 2\lambda \xi_7 - \lambda + 5}{4\lambda}, \\ K_7(\xi_7, \eta_7) &= \left(\frac{\gamma z_\alpha - z_\beta}{(1-\gamma) z_\beta} \right) \eta_7 + \frac{z_\beta^2 - 2\lambda \xi_7 - \lambda + 5}{4\lambda}. \end{aligned}$$

The values of ξ_7 and η_7 are solved from $H_7(\xi_7, \eta_7) = 0$ and $K_7(\xi_7, \eta_7) = 0$ so that the size is $\alpha + o(1/n_0)$ and the minimum power is $\beta + o(1/n_0)$.

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	$\alpha = .01$				$\alpha = .05$				$\alpha = .10$			
	$\beta = .6$	$\beta = .7$	$\beta = .8$	$\beta = .9$	$\beta = .6$	$\beta = .7$	$\beta = .8$	$\beta = .9$	$\beta = .6$	$\beta = .7$	$\beta = .8$	$\beta = .9$
$\gamma = .25$	3.508	3.828	4.231	4.811	2.683	2.968	3.343	3.907	2.258	2.526	2.880	3.428
	1.181	1.501	1.905	2.485	1.039	1.323	1.698	2.262	0.980	1.245	1.599	2.146
	1.200	1.139	1.062	1.025	0.583	0.589	0.573	0.589	0.343	0.396	0.420	0.465
	-0.351	-0.499	-0.676	-0.763	-0.112	-0.136	-0.170	-0.145	0.004	0.003	0.014	0.070
$\gamma = .50$	5.159	5.702	6.336	7.216	3.797	4.339	4.973	5.853	3.072	3.612	4.246	5.126
	2.833	3.376	4.009	4.890	2.152	2.694	3.328	4.208	1.790	2.331	2.965	3.845
	1.221	1.117	1.041	1.018	0.590	0.530	0.507	0.560	0.366	0.313	0.318	0.411
	-0.306	-0.550	-0.727	-0.778	-0.141	-0.241	-0.278	-0.191	-0.056	-0.125	-0.119	0.000

Table 1. Values of δ (top entry), R (2nd), x_0 (3rd) and y_0 (bottom)

Table 2. Simulation results on size (upper entry) and power (lower) of P_i , $i = 1, 2, 3$ when $\alpha = .05$ and $\gamma = .5$

β	P	$m = 10$				$m = 15$			
		$n_0 = 15$	40	90	200	$n_0 = 15$	40	90	200
.6	P_1	0.048	0.051	0.051	0.052	0.036	0.050	0.051	0.052
		0.593	0.592	0.596	0.597	0.596	0.592	0.596	0.595
	P_2^A	0.048	0.050	0.050	0.050	0.036	0.050	0.049	0.052
		0.593	0.594	0.586	0.598	0.596	0.592	0.597	0.597
	P_2^B	0.048	0.051	0.049	0.049	0.036	0.047	0.049	0.049
		0.594	0.592	0.592	0.600	0.596	0.595	0.594	0.596
	P_2^C	0.048	0.050	0.048	0.050	0.036	0.048	0.050	0.051
		0.594	0.598	0.594	0.598	0.597	0.595	0.593	0.596
	P_3^C	0.042	0.054	0.048	0.048	0.033	0.051	0.047	0.051
		0.592	0.595	0.594	0.599	0.598	0.593	0.596	0.600
	P_3^B	0.037	0.052	0.048	0.048	0.030	0.050	0.049	0.050
		0.594	0.594	0.590	0.597	0.601	0.598	0.593	0.598
	P_3^A	0.027	0.049	0.052	0.048	0.025	0.044	0.051	0.049
		0.602	0.597	0.593	0.596	0.606	0.600	0.593	0.596
.9	P_1	0.044	0.052	0.049	0.051	0.036	0.051	0.049	0.051
		0.903	0.895	0.897	0.900	0.914	0.895	0.897	0.900
	P_2^A	0.044	0.051	0.049	0.050	0.036	0.051	0.049	0.048
		0.903	0.892	0.894	0.900	0.914	0.894	0.896	0.898
	P_2^B	0.043	0.051	0.049	0.049	0.036	0.049	0.049	0.051
		0.902	0.889	0.894	0.899	0.914	0.896	0.895	0.894
	P_2^C	0.045	0.050	0.047	0.050	0.037	0.049	0.049	0.049
		0.900	0.894	0.899	0.897	0.913	0.895	0.897	0.901
	P_3^C	0.041	0.055	0.046	0.050	0.033	0.049	0.048	0.050
		0.904	0.891	0.897	0.896	0.917	0.894	0.897	0.897
	P_3^B	0.035	0.052	0.048	0.048	0.030	0.050	0.049	0.052
		0.910	0.890	0.894	0.900	0.918	0.894	0.896	0.896
	P_3^A	0.028	0.051	0.053	0.049	0.025	0.045	0.050	0.049
		0.927	0.894	0.894	0.900	0.929	0.901	0.897	0.900

β	P	$m = 10$				$m = 15$			
		$n_0 = 15$	40	90	200	$n_0 = 15$	40	90	200
.6	P_L	16.7(4.9)	41.1(9.8)	91.5(14.0)	201.6(20.2)	18.2(3.6)	41.3(9.4)	91.5(13.9)	201.7(20.1)
	P_2^A	16.7(4.9)	41.1(9.8)	91.8(14.1)	202.5(21.3)	18.2(3.6)	41.3(9.4)	91.6(13.9)	201.9(20.2)
	P_2^B	16.7(4.9)	41.2(9.8)	92.0(14.3)	202.7(21.3)	18.2(3.6)	41.3(9.4)	91.6(13.9)	201.9(20.4)
	P_2^C	16.8(5.0)	41.3(10.0)	92.0(14.2)	202.7(21.5)	18.2(3.6)	41.3(9.6)	91.7(13.9)	202.0(20.6)
	P_3^A	17.4(5.1)	41.1(10.8)	92.1(15.4)	203.0(22.1)	18.8(3.9)	41.4(10.2)	91.7(14.9)	202.2(21.1)
	P_3^B	18.3(5.3)	41.3(11.5)	91.6(17.3)	202.8(24.4)	19.6(4.2)	41.8(10.7)	91.5(16.7)	202.2(23.2)
	P_3^C	21.6(5.8)	43.3(13.2)	92.2(22.3)	203.1(34.1)	22.4(5.4)	44.4(11.7)	92.4(21.1)	202.8(31.2)
	P_{SL}	31.3(14.3)	81.6(38.1)	186.0(87.1)	407.4(189.6)	29.5(10.5)	77.0(29.0)	174.7(66.1)	381.6(143.4)
.9	P_L	16.7(4.9)	41.0(9.8)	91.5(13.9)	201.6(20.0)	18.1(3.6)	41.2(9.4)	91.5(13.8)	201.6(20.0)
	P_2^A	16.7(4.9)	41.0(9.8)	91.8(14.2)	202.5(21.3)	18.1(3.6)	41.2(9.4)	91.5(13.9)	201.8(20.4)
	P_2^B	16.7(4.9)	41.2(9.9)	91.9(14.3)	202.6(21.4)	18.1(3.6)	41.3(9.4)	91.6(13.8)	201.9(20.4)
	P_2^C	16.8(5.0)	41.2(10.0)	92.0(14.2)	202.7(21.4)	18.1(3.6)	41.3(9.5)	91.6(14.0)	202.0(20.5)
	P_3^A	17.3(5.0)	41.1(10.8)	92.0(15.5)	203.1(22.2)	18.7(3.9)	41.4(10.2)	91.7(15.0)	202.1(21.0)
	P_3^B	18.2(5.3)	41.3(11.4)	91.3(17.3)	203.0(24.2)	19.5(4.2)	41.9(10.7)	91.4(16.7)	202.1(23.5)
	P_3^C	21.5(5.8)	43.0(13.2)	92.1(22.1)	202.9(33.6)	22.2(5.3)	44.2(11.6)	92.2(21.1)	202.7(31.3)
	P_{SL}	24.1(10.9)	62.9(29.2)	141.6(66.9)	313.0(148.1)	22.8(7.4)	58.6(22.0)	130.2(49.2)	289.1(108.8)

Table 3. Simulation results on the average sample size (standard deviation)
when $\alpha = .05$ and $\gamma = .5$